

Software Test Plan (STP)

Project: Vehicle Parking System

Version: 1.0

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Status: Draft

Revision History

Version	Date	Change Summary
1.0	28-09-2026	Initial STP draft

Approvals

Role	Name	Signature / Date
QA Lead		
Dev Lead		
Course Coordinator		

Table of Contents

1. Introduction
2. Test Items
3. Features to be Tested
4. Features Not to be Tested
5. Test Approach / Strategy
6. Test Environment
7. Test Schedule
8. Test Deliverables
9. Roles and Responsibilities
10. Risks and Mitigation
11. Assumptions & Dependencies
12. Suspension & Resumption Criteria
13. Test Case Management & Traceability
14. Test Metrics & Reporting

1. Introduction

Purpose

This document defines the test plan for the Vehicle Parking System v1.0. It outlines test objectives, scope, strategy, resources, schedule, and responsibilities to ensure all functional and non-functional requirements defined in the SRS are verified before release.

Scope

Testing covers all VPS features: vehicle entry/exit, slot management, fee calculation, payment processing, reservation, and the admin portal. Hardware firmware, core payment gateway internals, and ALPR camera vendor software are excluded.

References

- VPS SRS v1.0 — docs/SRS/SRS_Vehicle_Parking_System.pdf
- VPS SAD v1.0 — docs/SAD/SAD_VehicleParkingSystem.pdf
- PCI-DSS v4.0 Standards
- WCAG 2.1 AA Guidelines

Definitions

Term	Definition
VPS	Vehicle Parking System
SRS	Software Requirements Specification
RTM	Requirements Traceability Matrix
UAT	User Acceptance Testing
TC	Test Case
LP	License Plate
ALPR	Automatic License Plate Recognition

2. Test Items

- Vehicle Entry module (LP capture, ticket generation, barrier control)
- Vehicle Exit module (ticket scan, fee display, barrier control)
- Slot Management module (real-time occupancy, categorisation)
- Fee Calculator module (duration-based and rate-schedule pricing)
- Payment Service module (card, cash, digital wallet)
- Reservation module (advance booking, grace period release)
- Admin Web Portal (dashboard, rate config, reports, user management)
- Auth Service (login, RBAC, account lockout)
- Logging & Monitoring module

3. Features to be Tested

Features mapped to SRS requirement IDs:

Feature	Req ID(s)
License plate capture at entry (ALPR + manual)	VPS-F-001
Slot assignment and occupancy marking	VPS-F-002
Ticket generation at entry	VPS-F-003
Real-time slot availability display	VPS-F-004
Slot categorisation (regular, disabled, EV)	VPS-F-005
Entry blocked when parking is full	VPS-F-006
Parking fee calculation	VPS-F-007

Feature	Req ID(s)
Payment processing at exit	VPS-F-008
Receipt generation and delivery	VPS-F-009
Slot marked available on exit	VPS-F-010
Advance slot reservation	VPS-F-011
Auto-release of expired reservation	VPS-F-012
Admin occupancy dashboard	VPS-F-013
Rate schedule configuration	VPS-F-014
Report export (CSV/PDF)	VPS-F-015
Transaction response time $\leq 5s$	VPS-NF-001
99.9% system availability	VPS-NF-002
PCI-DSS compliance	VPS-NF-003
5-year audit log retention	VPS-NF-004
WCAG 2.1 AA accessibility	VPS-NF-005
TLS 1.2+ on all connections	VPS-SR-001
Password complexity policy	VPS-SR-002
RBAC enforcement	VPS-SR-003
Account lockout after 5 failures	VPS-SR-004
AES-256 encryption at rest	VPS-SR-005

4. Features Not to be Tested

- Core payment gateway internal processing (vendor responsibility)
- ALPR camera firmware and image processing algorithms (vendor responsibility)
- Physical barrier hardware and motor control firmware
- Bank core settlement systems beyond the API interface
- Operating system and network infrastructure

5. Test Approach / Strategy

Test Levels

Level	Description
Unit Testing	Individual service/module functions tested in isolation (Fee Calculator, Slot Manager logic)
Integration Testing	Inter-service communication tested (Entry Terminal ↔ API Gateway ↔ Slot Manager ↔ DB)
System Testing	End-to-end VPS flows tested against all SRS requirements
Acceptance Testing (UAT)	Stakeholder sign-off testing against acceptance criteria in SRS

Test Types

Type	Description
Functional Testing	Verify all 15 FRs behave as specified
Regression Testing	Re-run test suite after every bug fix or new build
Performance Testing	Verify response time $\leq 5s$ at 90th percentile under load

Type	Description
Security Testing	Validate TLS, RBAC, account lockout, encryption, PCI-DSS
Usability Testing	Verify terminal UI clarity, accessibility (WCAG 2.1 AA)
Boundary Testing	Test edge cases: zero balance, max duration, full capacity

Entry Criteria

- Stable build delivered by development team
- Test environment configured and verified
- Test data (dummy vehicles, slots, accounts) available
- SRS and SAD reviewed and approved

Exit Criteria

- 100% of planned test cases executed
- 0 critical or high defects open
- All acceptance criteria in SRS satisfied
- RTM shows full coverage of requirements

5.1 Security Validation

- Validate PIN/password masking — no credentials in logs or UI
- TLS 1.2+ verification on all endpoints using network scan tools
- PCI-DSS compliance checks — no plaintext PAN in DB or logs
- Input fuzzing on LP entry field, ticket ID, and payment fields
- Penetration testing of authentication and admin portal flows
- RBAC boundary testing — attempt cross-role access and verify rejection
- Account lockout verification after 5 failed login attempts

6. Test Environment

Hardware

- Entry terminal kiosk with simulated barrier, LP camera, and ticket printer
- Exit terminal kiosk with simulated barrier, card reader, and receipt printer
- Admin workstation (standard browser)
- Load test server for JMeter-based performance testing

Software

Component	Version
VPS Application	v1.0 (build under test)
Database	MySQL 8.x (provisional assumption; confirm with implementation)
Payment Gateway	Sandbox / mock environment
ALPR Service	Mock service with pre-defined LP responses
OS	Ubuntu 22.04 LTS (server), Windows 11 (admin workstation)

Tools

Tool	Purpose
Selenium	UI automation for terminal and admin portal
Postman / Newman	API testing for /entry, /exit, admin endpoints
JMeter	Performance and load testing
OWASP ZAP	Security / penetration testing
Jira	Defect tracking and test management
JUnit 5	Unit testing (only if the implemented backend uses Java; confirm stack first)

Test Data

- 50 dummy vehicle records with various license plates
- 3 parking facility configurations (small, medium, large capacity)
- Multiple rate schedules (standard, peak, flat-rate)
- Test payment cards (sandbox credentials for all payment methods)
- Admin, attendant, and customer user accounts

7. Test Schedule

Milestone	Target Date
Test case design complete	05-10-2026
Test environment setup verified	07-10-2026
Unit & integration test execution	08-10-2026 to 12-10-2026
System test execution	13-10-2026 to 20-10-2026
Security & performance testing	21-10-2026 to 23-10-2026
UAT	24-10-2026 to 27-10-2026
Test summary report	28-10-2026

8. Test Deliverables

- Test Plan (this document)
- Test Cases (manual and automated)
- Test Scripts (Selenium, Postman collections, JMeter plans)
- Test Data sets
- Test Execution Logs
- Defect Reports (Jira)
- Final Test Summary Report

9. Roles and Responsibilities

Role	Responsibility
QA Lead	Prepare test plan, coordinate execution, sign off
Test Engineer	Design and execute test cases, log defects
Developer	Support defect triage and fixes

Role	Responsibility
Product Owner / Instructor	Review results, approve UAT sign-off

10. Risks and Mitigation

Risk	Mitigation
Delay in stable build delivery	Request early smoke builds; run smoke test immediately on receipt
Test environment downtime	Maintain backup environment on cloud VM
ALPR vendor mock not available	Build internal LP mock service using pre-defined inputs
Payment gateway sandbox instability	Use fully offline mock payment service as fallback
Incomplete test data	Prepare and version-control all test data sets in advance

11. Assumptions & Dependencies

- Payment gateway sandbox will be stable and available throughout testing
- ALPR mock service will be ready before system test execution begins
- Test data (vehicles, accounts, rate schedules) will be prepared before execution
- Hardware drivers and terminal simulators will be provided before integration testing

12. Suspension & Resumption Criteria

Suspend testing if:

- Test environment is unavailable for more than 4 consecutive hours
- A build blocks more than 30% of planned test cases
- A critical security vulnerability is discovered during testing

Resume testing if:

- Blocking defects are resolved and verified fixed
- Test environment is stabilised
- A new build is delivered with confirmed fixes

13. Test Case Management & Traceability

The Requirements Traceability Matrix (RTM) maps SRS requirements to test case IDs in docs/TestCases/Vehicle_Parking_System_Test_Cases.xlsx.

Test ID convention: The spreadsheet's canonical test case identifiers are TC-001 through TC-040. The test case workbook records planned tests only; all cases remain **Not Executed** until run against an implementation or suitable test environment.

Req ID	Requirement Short	Test Case(s)	Traceability Notes
VPS-F-001	Capture license plate	TC-001, TC-002, TC-003, TC-036	Automatic capture, manual entry, validation case, and entry integration. TC-003 validation rules need confirmation.
VPS-F-002	Assign slot and mark occupied	TC-004, TC-005, TC-036, TC-038	Allocation, occupied-slot protection, entry integration, and concurrent allocation.
VPS-F-003	Generate entry ticket	TC-006, TC-036	Ticket details and entry integration.
VPS-F-004	Display real-time slot availability	TC-007, TC-040	Availability display and capacity-boundary state.
VPS-F-005	Categorise parking slots	TC-008	EV-eligible slot assignment; admin configuration coverage may need a separate case.

Req ID	Requirement Short	Test Case(s)	Traceability Notes
VPS-F-006	Block entry when parking is full	TC-009, TC-040	Full-capacity rejection and final-slot boundary.
VPS-F-007	Calculate parking fee	TC-010, TC-011, TC-037	Standard fee, rate boundary, and exit integration. Concrete rate schedule and rounding rules need definition.
VPS-F-008	Process payment at exit	TC-012, TC-013, TC-014, TC-037	Successful payment, failed payment, gateway failure, and exit integration.
VPS-F-009	Generate and deliver receipt	TC-015, TC-037	Receipt content and exit integration.
VPS-F-010	Mark slot available on exit	TC-016, TC-037	Slot release and exit integration. TC-037 verifies this only if the final slot state is observed.
VPS-F-011	Advance reservation	TC-018, TC-019	Reservation creation and overlapping reservation conflict; overlap rules need confirmation.
VPS-F-012	Auto-release expired reservation	TC-020	Automatic release after grace-period expiry.
VPS-F-013	Admin occupancy dashboard	TC-021	Dashboard case. TC-022 is not counted as coverage unless history search is confirmed in the approved SRS.
VPS-F-014	Configure rate schedule	TC-023, TC-024	Rate configuration and invalid-rate handling; invalid-input rules need definition.
VPS-F-015	Export reports	TC-025, TC-039	Export format/content and non-admin export denial, provided report export is in the approved SRS.
VPS-NF-001	Transaction response time ≤ 5 seconds	TC-026	Define transaction mix, concurrency, data volume, duration, environment, and measurement method.
VPS-NF-002	99.9% system availability	TC-027	Short test windows cannot establish monthly availability; use a defined observation period and supporting evidence.
VPS-NF-003	PCI-DSS compliance	TC-028	Limited check for plaintext payment data only; this is not proof of full PCI-DSS compliance.
VPS-NF-004	5-year audit-log retention	TC-029	Configuration evidence only unless actual retention behaviour can be demonstrated.
VPS-NF-005	WCAG 2.1 AA accessibility	TC-030	Define the pages, criteria, tools, and evidence; a limited manual check does not prove full conformance.
VPS-SR-001	TLS 1.2+ on connections	TC-031	Test all applicable endpoints, not only a single endpoint.
VPS-SR-002	Password complexity policy	TC-032	Obtain explicit policy rules and test examples before execution.
VPS-SR-003	RBAC enforcement	TC-033, TC-039	Role permission checks and report-export authorization, if export is in scope.
VPS-SR-004	Account lockout after 5 failed attempts	TC-034	Lockout behaviour; duration and recovery behaviour need clarification if required.
VPS-SR-005	AES-256 encryption at rest	TC-035	Verify evidence of the specified AES-256 configuration; a generic protected-at-rest setting is insufficient.

Traceability items requiring clarification

- **TC-017 — Invalid parking ticket at exit:** The SRS describes ticket validation generally but does not provide a dedicated requirement ID or explicit acceptance criterion for invalid-ticket rejection. Do not map it to VPS-F-010, which concerns slot release. Keep the case pending clarification or approval as an additional test.
- **TC-003 — Invalid/empty plate:** Confirm the accepted license-plate format and expected behaviour for blank input.
- **TC-008 — Vehicle-category slot assignment:** Confirm whether admin configuration of slot categories requires a separate test.
- **TC-010/TC-011/TC-023/TC-024 — Rate handling:** Define concrete rates, rounding, boundary inclusivity, and invalid schedule rules before execution.
- **TC-019 — Reservation conflicts:** Confirm the rule for overlapping reservation windows.
- **TC-022 — Vehicle/parking history search:** Do not count this as required coverage until its inclusion in the approved SRS is confirmed.
- **TC-026 to TC-030 — Non-functional requirements:** Confirm the measurement plan and evidence required for performance, availability, payment-data handling, retention, and accessibility.
- **TC-032/TC-034 — Authentication policy details:** Confirm password rules and account-lockout duration/recovery behaviour where the SRS does not specify them.

- **TC-036 to TC-040 — Integration, concurrency, and boundary cases:** Treat these as supplementary scenarios. Count them toward a requirement only when their steps and expected results explicitly verify that requirement.
- **Coverage status:** A mapping indicates planned coverage, not successful verification. A requirement should be reported as verified only after relevant tests have been executed, passed, and supported by recorded evidence.

14. Test Metrics & Reporting

Metrics Collected

- % test cases executed
- % test cases passed / failed
- Defect density (defects per module)
- Defect severity distribution (critical / high / medium / low)
- Defect aging (average days open)
- Requirement coverage (% requirements with passing test cases)

Reports

- Daily test execution status update
- Weekly defect summary
- Final Test Summary Report (post-UAT)